



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Human-AI Workforce with Autonomous AI Agents

TGS Ref No: TGS-2024043854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.9

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 July 2026	Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).	Dr. Alfred Ang
1.1	14 July 2026	Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.	Dr. Alfred Ang
1.2	14 July 2026	Visual redesign: all content slides card-format; YouTube links removed from the deck and Learner Guide (videos remain in the labs); live Hermes screenshots and skill-category grids added; Hermes install lab expanded with the official install methods.	Dr. Alfred Ang
1.3	14 July 2026	Deck restyled in the dark masterclass theme end-to-end; added the Hermes masterclass slides (What is Hermes; The Bet: Compounding Value; platform matrix; Top-10 slash commands; Telegram setup; memory L1-vs-L2 and session-search; what-is-a-tool, built-in tool catalog, tools-in-action, cron scheduling, delegation, kanban orchestration and OpenClaw topic slides); Lab 13 rebuilt as the multi-agent video team (profiles + kanban) with detailed Labs 11-14 guides; fade slide transitions and the live Achievements	Dr. Alfred Ang

		dashboard screenshot.	
1.4	15 July 2026	Security masterclass slides added to the Hermes Security lab (defense-in-depth layers; the gateway's six-check default-deny trust chain; channel lockdown dials; the three approval modes + the Tirith scanner; a live Dangerous-Command approval; YOLO mode; the unrecoverable-command blocklist floor; prompt-injection scanning of context files). Security lab steps expanded with the channel allowlist and approvals.mode.	Dr. Alfred Ang
1.5	15 July 2026	Sixteen concept diagrams imported from the original Mastering OpenClaw masterclass deck into Topic 2 (OpenClaw OS control plane; deployment; hub-and-spoke gateway; tools vs skills; SKILL-on-demand; 6-phase execution loop; exec; HEARTBEAT + autonomic tick; workspace file persistence; context assembly; compaction; defense methods; poisoned-skill anatomy; sessions_spawn; sub-agent routing).	Dr. Alfred Ang
1.6	15 July 2026	Topic 3 (Paperclip) rebuilt as the 'Tertiary AI News Research' company across 6 labs (27-32): install on Windows & Mac; company, CEO & mission; model adaptors; a detailed task backlog whose core task hires the team; the Tavily Search API via Secrets; and hiring the members under the	Dr. Alfred Ang

		hiring task — with live dashboard screenshots. Natural-language prompt steps added to the Hermes and OpenClaw labs alongside the CLI. Platform logos added to the Three Platforms page. 32 labs total.	
1.7	15 July 2026	Sixty-eight content slides migrated from the original 'Mastering OpenClaw' masterclass deck into Topic 2 as framed figures in their matching labs — agent evolution and Agent = Model + Harness, the OpenClaw Wikipedia history and OpenAI acquisition, deployment options and services, model stacks and Mission Control, channels, skills and ClawHub risks, tools (AgentMail, Agent Browser, Firecrawl), system prompt / workspace / context and memory, security lockdown, slash commands, crons, and use cases (NFT certificates, dashboard, cost saving, Google integration).	Dr. Alfred Ang
1.8	15 July 2026	Imported masterclass figures re-blended into the dark deck theme: original title and copyright strips cropped, white canvases converted to the deck background with text inverted to ivory (embedded screenshots untouched), frameless rendering, and duplicated titles reworded.	Dr. Alfred Ang
1.9	16 July 2026	Deck curated per trainer review: the bulk masterclass figure imports were removed,	Dr. Alfred Ang

		keeping nine key figures (Wikipedia history, Agent=Model+Harness , SKILL.md, ClawHub, AgentMail, 10-steps lockdown, HEARTBEAT, autonomic tick, workspace persistence) re-exported at high resolution, with the Topic 2 opening story reordered ahead of Key Concepts. Every CLI step now also shows the natural-language prompt alternative ('Or just ask') across the Hermes and OpenClaw labs, in the deck and Learner Guide.	
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Course Information

Course Title	Build a Human-AI Workforce with Autonomous AI Agents
WSQ Course Reference	TGS-2024043854
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on autonomous-AI-agent labs (local machine + Docker)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Hermes Agent (Topic 1) & OpenClaw (Topic 2)

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)
9:50–10:30	40 min	Topic 1 — Hermes Agent — Your Autonomous Personal Chief-of-Staff: what an autonomous AI agent is, the agent stack (model, memory, skills, tools) and human-in-the-loop governance

		(concepts + demo)
10:30–11:00	30 min	Hands-on: Lab 1: Install and Setup Hermes; Lab 2: Deployment — Local Install & Hermes Desktop App; Lab 3: Memory and Plugins
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 4: Skills; Lab 5: Providers & Model; Lab 6: MCP and Tools; Lab 7: Cron Jobs & Automation; Lab 8: Subagents & Delegation; Lab 9: Profile and Kanban
13:00–14:00	60 min	Lunch break
14:00–14:40	40 min	Hands-on: Lab 10: Security; Lab 11: Use Case — Make a Video with Hyperframe; Lab 12: Visualization; Lab 13: Build a Multi-Agent Video Team (Profiles + Kanban); Lab 14: Webhook
14:40–15:20	40 min	Topic 2 — OpenClaw — Automating a Business Back-Office: the OpenClaw gateway daemon, providers, channels, skills, tools, dreaming and crons for a business back-office (concepts + demo)
15:20–15:35	15 min	Tea break
15:35–18:15	160 min	Hands-on: Lab 15: Install OpenClaw; Lab 16: Models & Providers; Lab 17: Channels; Lab 18: Skills; Lab 19: Tools & Integrations; Lab 20: Commands; Lab 21: Cron Jobs & Heartbeat; Lab 22: Memory & Context; Lab 23: Security; Lab 24: Multi-Agent Workforce; Lab 25: Dreaming — Idle Reflection & Self-Improvement; Lab 26: Use Cases & Key Functions
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Paperclip (Topic 3) & Final Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:30	45 min	Topic 3 — Paperclip — Running & Governing a Company of AI Agents: Docker-hosted company of agents (Tertiary AI News Research), model adaptors,

		the mission, a detailed task backlog, Tavily live search, and hiring behind human approval gates (concepts + demo)
10:30–11:00	30 min	Hands-on: Lab 27: Install Paperclip on Windows & Mac; Lab 28: Setup Company, CEO & Mission
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 29: Setup Adaptor; Lab 30: Create the Task Backlog; Lab 31: Add the Tavily Search API; Lab 32: Hire the Members Under the Hiring Task
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: complete and extend Labs 27–32 — run the news desk end-to-end (backlog → Tavily research → briefing review). Course recap and next steps
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Consolidation, Q&A and assessment preparation
16:15–16:30	15 min	Briefing for Assessment
16:30–17:30	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
17:30–18:30	60 min	Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic / Platform	Weighting	Labs
Topic 01: Hermes Agent — Your Autonomous Personal Chief-of-Staff	33%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14
Topic 02: OpenClaw — Automating a Business Back-Office	33%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26
Topic 03: Paperclip — Running & Governing a Company of AI Agents	34%	Lab 27, Lab 28, Lab 29, Lab 30, Lab 31, Lab 32